DESCENT OF THE EXISTENCE OF A CONNECTED COMPONENT MEETING EVERY FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 13108-13190, immediately following the statement at 13087-13106.

Editorial note. Nine cross-chapter pointers (Morphisms, Descent) are recorded in assembly.external_references; the descent step that uses them is displayed in the span.

Editorial note. The proof contains one 'details omitted' phrase for the quasi-compactness reduction; the flagging note appended below gives its source line.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 13188 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- The situation of connected components along a section. chapter situation situation-connected-along-section, together with lemmas lemma-connected-along-section-open and lemma-connected-along-section-locally-constructible, cited throughout. Provides the open subscheme X_Z^0 of points connected to the section inside their fibre and its local constructibility.
- Base change of connectedness along a section. chapter lemma lemma-base-change-connected-along-section, cited inside the proof. Shows the construction is compatible with base change, which is what the descent argument tests on the two projections.
- Flatness, finite presentation and finite locally freeness are fppf local. Morphisms lemmas cited in the proof (base change and composition of flat, finite presentation and finite locally free morphisms). Lets the hypotheses be checked after base change along the fppf covering Z -> Y.
- Openness of an fppf covering and descent of open immersions. Morphisms, lemma morphisms-lemma-fppf-open and Descent, lemma descent-lemma-open-fpqc-covering, cited at proof lines 13181-13183. Descends the open subscheme X^0 of X_Z to an open of X.

Source reference. Original reference 03N0: focused-seeds/geometry/S9-morph-descent-connected-fibres/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 055M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8187–8187. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 055P: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8268–8268. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 055R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8387–8387. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02KD: focused-seeds/geometry/S9-morph-descent-connected-fibres/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01TS: focused-seeds/geometry/S9-morph-descent-connected-fibres/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01U9: focused-seeds/geometry/S9-morph-descent-connected-fibres/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02KC: focused-seeds/geometry/S9-morph-descent-connected-fibres/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01TR: focused-seeds/geometry/S9-morph-descent-connected-fibres/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01U7: focused-seeds/geometry/S9-morph-descent-connected-fibres/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01WN: focused-seeds/geometry/S9-morph-descent-connected-fibres/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01UA: focused-seeds/geometry/S9-morph-descent-connected-fibres/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 055L: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8179–8179. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Consider a diagram of morphisms of schemes*

$$\begin{array}{ccc} Z & \xrightarrow{\sigma} & X \\ & \searrow & \downarrow \\ & & Y \end{array}$$

an a point $y \in Y$. Assume

(1) *$X \to Y$ is of finite presentation and flat,*
(2) *$Z \to Y$ is finite locally free,*
(3) *$Z_y \neq \emptyset$,*
(4) *all fibres of $X \to Y$ are geometrically reduced, and*
(5) *X_y is geometrically connected over $\kappa(y)$.*

Then there exists a quasi-compact open $X^0 \subset X$ such that $X^0_y = X_y$ and such that all nonempty fibres of $X^0 \to Y$ are geometrically connected.

Proof. In this proof we will use that flat, finite presentation, finite locally free are properties that are preserved under base change and composition. We will also use that a finite locally free morphism is both open and closed. You can find these facts as Morphisms, Lemmas 01U9, 01TS, 02KD, 01U7, 01TR, 02KC, 01UA, and 01WN.

Note that $X_Z \to Z$ is flat morphism of finite presentation which has a section s coming from σ. Let X^0_Z denote the subset of X_Z defined in Situation 055L. By Lemma 055R it is an open subset of X_Z.

The pullback $X_{Z\times_Y Z}$ of X to $Z \times_Y Z$ comes equipped with two sections s_0, s_1, namely the base changes of s by $\mathrm{pr}_0, \mathrm{pr}_1 : Z \times_Y Z \to Z$. The construction of Situation 055L gives two subsets $(X_{Z\times_Y Z})^0_{s_0}$ and $(X_{Z\times_Y Z})^0_{s_1}$. By Lemma 055M these are the inverse images of X^0_Z under the morphisms $1_X \times \mathrm{pr}_0, 1_X \times \mathrm{pr}_1 : X_{Z\times_Y Z} \to X_Z$. In particular these subsets are open.

Let $(Z \times_Y Z)_y = \{z_1, \ldots, z_n\}$. As X_y is geometrically connected, we see that the fibres of $(X_{Z\times_Y Z})^0_{s_0}$ and $(X_{Z\times_Y Z})^0_{s_1}$ over each z_i agree (being equal to the whole fibre). Another way to say this is that

$$s_0(z_i) \in (X_{Z\times_Y Z})^0_{s_1} \quad \text{and} \quad s_1(z_i) \in (X_{Z\times_Y Z})^0_{s_0}.$$

Since the sets $(X_{Z\times_Y Z})^0_{s_0}$ and $(X_{Z\times_Y Z})^0_{s_1}$ are open in $X_{Z\times_Y Z}$ there exists an open neighbourhood $W \subset Z \times_Y Z$ of $(Z \times_Y Z)_y$ such that

$$s_0(W) \subset (X_{Z\times_Y Z})^0_{s_1} \quad \text{and} \quad s_1(W) \subset (X_{Z\times_Y Z})^0_{s_0}.$$

Then it follows directly from the construction in Situation 055L that

$$p^{-1}(W) \cap (X_{Z\times_Y Z})^0_{s_0} = p^{-1}(W) \cap (X_{Z\times_Y Z})^0_{s_1}$$

where $p : X_{Z\times_Y Z} \to Z \times_W Z$ is the projection. Because $Z \times_Y Z \to Y$ is finite locally free, hence open and closed, there exists an affine open neighbourhood $V \subset Y$ of y such that $q^{-1}(V) \subset W$, where $q : Z \times_Y Z \to Y$ is the structure morphism. To

prove the lemma we may replace Y by V. After we do this we see that $X_Z^0 \subset Y_Z$ is an open such that

$$(1_X \times \mathrm{pr}_0)^{-1}(X_Z^0) = (1_X \times \mathrm{pr}_1)^{-1}(X_Z^0).$$

This means that the image $X^0 \subset X$ of X_Z^0 is an open such that $(X_Z \to X)^{-1}(X^0) = X_Z^0$, see Descent, Lemma 03N0. Finally, X^0 is quasi-compact because X_Z^0 is quasi-compact by Lemma 055P (use that at this point Y is affine, hence X is quasi-compact and quasi-separated, hence locally constructible is the same as constructible and in particular quasi-compact; details omitted). In this way we see that X^0 has all the desired properties. $\square$